

XAudio2Player Sample 4 – Architecture Diagram

This diagram describes the high-level architecture of the XAudio2Player Sample 4. It shows how Media Foundation, XAudio2, the worker thread, and the GUI interact in a streaming, multi-buffer playback model.

GUI Thread	Worker Thread	Media Foundation	XAudio2
User input Play / Pause / Stop / Seek Progress bar updates	Command processing Buffer refill State polling	IMFSourceReader Streaming decode Sample-accurate seek	SourceVoice Multi-buffer playback Effects processing

Data Flow

1. The GUI thread sends commands (Play, Pause, Stop, Seek) to the worker thread.
2. The worker thread controls playback state and requests decoded audio data.
3. Media Foundation decodes compressed audio incrementally into PCM buffers.
4. XAudio2 consumes PCM buffers using a multi-buffer streaming model.
5. Playback state is polled and reported back to the GUI using thread-safe callbacks.

Key Design Principles

- Streaming decode instead of full-file buffering
- Dedicated worker thread for real-time audio processing
- Thread-safe state updates without message pumping
- Accurate sample-based timing and seeking
- Scalable to very large audio files